

# Lattice Point Counting via Commutative Algebra

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# 1 Lecture 1- May 25, 2026

## Semigroups from Linear Algebra

Let  $A$  be a matrix with rational or integral entries. After clearing denominators, every rational matrix relevant to lattice-point questions may be replaced by an integral matrix.

**Definition 1.1** (Semigroup). A *semigroup* is a set  $S$  equipped with an associative binary operation. In the context of this lecture, we primarily consider additive subsemigroups of  $\mathbb{Z}^n$ , which are subsets  $S \subseteq \mathbb{Z}^n$  that are closed under addition. That is, if  $x, y \in S$ , then  $x + y \in S$ .

The basic objects in the lecture are subsets of  $\mathbb{Z}^n$  cut out by linear equalities and inequalities. For example, the set of solutions

$$\{x \in \mathbb{Z}^n \mid Ax = 0\}$$

forms an additive semigroup. Similarly, the set of nonnegative solutions

$$\{x \in \mathbb{Z}^n \mid Ax \geq 0\}$$

is a semigroup.

**Definition 1.2** (Integral span). Let  $v_1, \dots, v_\ell \in \mathbb{Z}^n$ . The integral span of these vectors is

$$\text{span}_{\mathbb{Z}}\{v_1, \dots, v_\ell\} = \left\{ \sum_{i=1}^{\ell} \lambda_i v_i \mid \lambda_i \in \mathbb{Z} \right\}.$$

**Definition 1.3** (Positive span). Let  $v_1, \dots, v_\ell \in \mathbb{Z}^n$ . The positive span, or rational polyhedral cone generated by these vectors, is

$$\text{pos}\{v_1, \dots, v_\ell\} = \left\{ \sum_{i=1}^{\ell} \lambda_i v_i \mid \lambda_i \geq 0 \right\}.$$

**Example 1.4.** The set

$$\{x \in \mathbb{Z}^n \mid Ax = 0\}$$

is an integral span, and

$$\{x \in \mathbb{Z}^n \mid Ax \geq 0\}$$

is the lattice points of a positive span.

## Affine Semigroups

**Definition 1.5** (Affine semigroup). An *affine semigroup* is a finitely generated subsemigroup

$$S \subseteq (\mathbb{Z}^n, +).$$

That is, we can write

$$S = \mathbb{N}\{v_1, \dots, v_\ell\} := \left\{ \sum_{i=1}^{\ell} n_i v_i \mid n_i \in \mathbb{N} \right\}.$$

Here  $v_i \in \mathbb{Z}^n$ . Throughout these notes,  $\mathbb{k}$  denotes an algebraically closed field of characteristic 0.

**Definition 1.6** (Group generated by a semigroup). Let  $S$  be a semigroup. The group generated by  $S$  is

$$\mathbb{Z}S = \{s_1 - s_2 \mid s_1, s_2 \in S\}.$$

If  $S = \mathbb{N}\{v_1, \dots, v_\ell\}$ , then  $\mathbb{Z}S = \text{span}_{\mathbb{Z}}\{v_1, \dots, v_\ell\}$ .

**Definition 1.7** (Rank). The rank of an affine semigroup  $S$  is the rank of the abelian group  $\mathbb{Z}S$ :

$$\text{rank}(S) = \text{rank}_{\mathbb{Z}}(\mathbb{Z}S).$$

If  $\mathbb{Z}S = \mathbb{Z}^n$ , then  $\text{rank}(S) = n$ .

**Definition 1.8** (Saturation). Let  $S$  be an affine semigroup. We say that  $S$  is *saturated* if, whenever  $x \in \mathbb{Z}S$  and  $k \in \mathbb{N}_{>0}$  satisfy

$$kx \in S,$$

then  $x \in S$ . Equivalently,  $S$  is saturated if

$$S = \{x \in \mathbb{Z}S \mid kx \in S \text{ for some } k \in \mathbb{N}_{>0}\}.$$

*Remark 1.9.* Informally, a saturated semigroup has no missing lattice points inside its rational cone.

## Numerical Semigroups and the Frobenius Problem

**Definition 1.10** (Numerical semigroup). A numerical semigroup is a subsemigroup  $S \subseteq \mathbb{N}$  such that  $\mathbb{N} \setminus S$  is finite.

**Example 1.11.** Let

$$S = \mathbb{N}\{a, b\} = \{ma + nb \mid m, n \in \mathbb{N}\} \subseteq \mathbb{N}.$$

Assume  $\gcd(a, b) = 1$ . If  $\gcd(a, b) \neq 1$ , then  $S$  generates a proper subgroup of  $\mathbb{Z}$ , so infinitely many natural numbers are missing from  $S$ . Conversely, if  $\gcd(a, b) = 1$ , the classical Frobenius theorem implies that all sufficiently large natural numbers lie in  $S$ .

**Theorem 1.12** (Sylvester's theorem). Let  $a, b \in \mathbb{N}_{>0}$  satisfy  $\gcd(a, b) = 1$ , and let  $S = \mathbb{N}\{a, b\}$ . Define  $f(a, b) = (a - 1)(b - 1)$ .

(1) If  $N \geq f(a, b)$ , then

$$N \in S.$$

The largest integer not in  $S$  is  $f(a, b) - 1 = ab - a - b$ .

(2) The number of gaps in

$$\{0, 1, 2, \dots, f(a, b)\}$$

is exactly  $\frac{f(a, b)}{2}$ .

Equivalently, the Frobenius number of  $S$  is

$$F(a, b) = ab - a - b.$$

**Example 1.13.** Let

$$S = \mathbb{N}\{3, 5\} \subseteq \mathbb{N}.$$

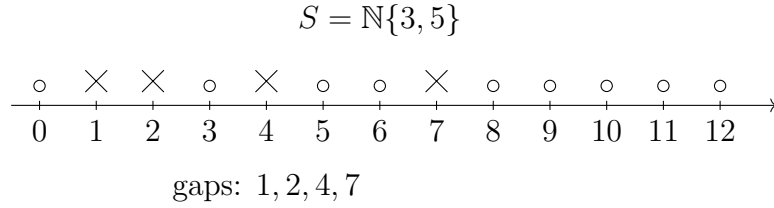
We can compute the elements of  $S$ :

$$S = \{0, 3, 5, 6, 8, 9, 10, 11, 12, \dots\}.$$

The set of gaps is:

$$\mathbb{N} \setminus S = \{1, 2, 4, 7\}.$$

The Frobenius number is  $F(3, 5) = 3 \cdot 5 - 3 - 5 = 7$ .



**Question 1.14** (Three-generator Frobenius problem). Let

$$S = \mathbb{N}\{a, b, c\} \subseteq \mathbb{N}$$

where  $\gcd(a, b, c) = 1$ . What is  $F(a, b, c)$ ? For example, consider  $S = \mathbb{N}\{9, 10, 11\}$ . There is no general simple formula for the Frobenius number in three generators.

## Examples and Nonexamples of Saturation

**Example 1.15.** Let

$$S = \mathbb{N}\{3, 5\} \subseteq \mathbb{N}.$$

Because  $\mathbb{N} \setminus S = \{1, 2, 4, 7\}$ ,  $S$  is not saturated inside  $\mathbb{N}$ .

**Example 1.16.** Let

$$S = \mathbb{N}\{a, b\} \subseteq \mathbb{N}.$$

If  $d = \gcd(a, b) \neq 1$ , then  $\mathbb{Z}S = d\mathbb{Z}$ , so every nonnegative integer not divisible by  $d$  is missing from  $S$ .

**Example 1.17** (A nonsaturated affine semigroup). Let

$$S = \mathbb{N}\{(1, 0), (1, 2), (2, 1)\} \subseteq \mathbb{Z}^2.$$

The cone generated by  $S$  is the same cone generated by  $(1, 0)$  and  $(1, 2)$ , namely

$$\text{Cone}(S) = \{(x, y) \in \mathbb{R}^2 \mid 0 \leq y \leq 2x\}.$$

Unlike the semigroup  $\mathbb{N}\{(1, 0), (1, 2)\}$ , this semigroup generates the whole lattice  $\mathbb{Z}^2$ . Indeed,

$$\det \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = 1,$$

so the two elements  $(1, 0)$  and  $(2, 1)$  already generate  $\mathbb{Z}^2$  as a group. Hence

$$\mathbb{Z}S = \mathbb{Z}^2.$$

We now show that  $S$  is not saturated. Consider the lattice point

$$(1, 1) \in \mathbb{Z}S = \mathbb{Z}^2.$$

This point is not in  $S$ . Indeed, if

$$(1, 1) = a(1, 0) + b(1, 2) + c(2, 1)$$

with  $a, b, c \in \mathbb{N}$ , then comparing coordinates gives

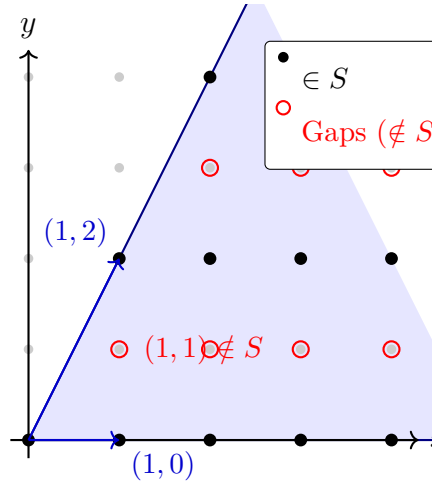
$$a + b + 2c = 1, \quad 2b + c = 1.$$

The second equation forces either  $b = 0, c = 1$ , but then the first equation gives  $a + 2 = 1$ , impossible. Hence  $(1, 1) \notin S$ .

However,

$$2(1, 1) = (2, 2) = (1, 0) + (1, 2) \in S.$$

Thus there exists an element  $x = (1, 1) \in \mathbb{Z}S$  such that  $x \notin S$ , but  $2x \in S$ . Therefore  $S$  is not saturated.



## Pointed Cones and Gordan's Lemma

A central source of affine semigroups is given by lattice points inside rational polyhedral cones.

**Definition 1.18** (Pointed cone). Let  $\sigma$  be a cone. We say that  $\sigma$  is *pointed* if

$$\sigma \cap (-\sigma) = \{0\}.$$

Geometrically, this says that the cone has its apex at 0.

**Example 1.19.** Let

$$\sigma = \{x \in \mathbb{R}^n \mid Ax \geq 0\}$$

be a pointed rational polyhedral cone. Then

$$S = \sigma \cap \mathbb{Z}^n$$

is a semigroup.

**Proposition 1.20.** *Let  $\sigma$  be a rational polyhedral cone, and set*

$$S = \sigma \cap \mathbb{Z}^n.$$

*If  $\mathbb{Z}S = \mathbb{Z}^n$ , then  $S$  is saturated.*

*Proof.* Suppose  $x \in \mathbb{Z}S$  and  $kx \in S$  for some  $k \in \mathbb{N}_{>0}$ . Since  $S \subseteq \sigma$ , we have

$$kx \in \sigma.$$

Since  $\sigma$  is a cone, it is closed under multiplication by positive real scalars. Hence

$$x = \frac{1}{k}(kx) \in \sigma.$$

Therefore  $x \in \sigma \cap \mathbb{Z}S$ . In the case  $\mathbb{Z}S = \mathbb{Z}^n$ , this is simply

$$x \in \sigma \cap \mathbb{Z}^n = S.$$

□

**Theorem 1.21** (Gordan's lemma). *Let*

$$\sigma = \text{pos}\{v_1, \dots, v_\ell\} \subseteq \mathbb{R}^n$$

*be a rational cone, where  $v_1, \dots, v_\ell \in \mathbb{Z}^n$ . Then*

$$S = \sigma \cap \mathbb{Z}^n$$

*is an affine semigroup (it is finitely generated).*

*Proof.* Let  $v \in S$ . Since  $v \in \sigma$ , there exist real numbers  $\lambda_i \geq 0$  such that

$$v = \sum_{i=1}^{\ell} \lambda_i v_i.$$

Write  $\lambda_i = \lfloor \lambda_i \rfloor + \{\lambda_i\}$ , where

$$\lfloor \lambda_i \rfloor \in \mathbb{N}, \quad 0 \leq \{\lambda_i\} < 1.$$

Then

$$v = \sum_{i=1}^{\ell} \lfloor \lambda_i \rfloor v_i + \sum_{i=1}^{\ell} \{\lambda_i\} v_i.$$

Define the fundamental parallelepiped

$$\text{Par}(v_1, \dots, v_\ell) = \left\{ \sum_{i=1}^{\ell} \mu_i v_i \mid 0 \leq \mu_i < 1 \right\}.$$

Since  $v$  and  $\sum_{i=1}^{\ell} [\lambda_i] v_i$  are lattice points, the second summand is also a lattice point. Hence

$$\sum_{i=1}^{\ell} \{\lambda_i\} v_i \in \text{Par}(v_1, \dots, v_{\ell}) \cap \mathbb{Z}^n.$$

The intersection  $\text{Par}(v_1, \dots, v_{\ell}) \cap \mathbb{Z}^n$  is a bounded discrete set, hence finite. It follows that every element of  $S$  is generated by the finite set

$$\{v_1, \dots, v_{\ell}\} \cup \left( \text{Par}(v_1, \dots, v_{\ell}) \cap \mathbb{Z}^n \right).$$

□

*Remark 1.22.* The lecture conclusion was:

$$\text{generators of } S \subseteq \{v_1, \dots, v_{\ell}\} \cup \left( \text{Par}(v_1, \dots, v_{\ell}) \cap \mathbb{Z}^n \right),$$

which proves the set is finitely generated.

**Definition 1.23** (Hilbert basis). Let  $S$  be a positive affine semigroup. A minimal set of semigroup generators for  $S$  is called the Hilbert basis of  $S$ , denoted  $\text{Hilb}(S)$ . For a positive affine semigroup, an element  $g \in S \setminus \{0\}$  belongs to  $\text{Hilb}(S)$  if and only if it cannot be written as

$$g = v + v'$$

with nonzero  $v, v' \in S$ .

## Semigroup Rings

Let  $S$  be an affine semigroup.

**Definition 1.24** (Semigroup ring). The semigroup ring of  $S$  over  $\mathbb{k}$  is

$$\mathbb{k}[S] = \mathbb{k}[t^m \mid m \in S].$$

For  $m = (m_1, \dots, m_n) \in \mathbb{Z}^n$ , we write

$$t^m = t_1^{m_1} \dots t_n^{m_n}.$$

The multiplication rule is  $t^m t^{m'} = t^{m+m'}$ .

*Remark 1.25.* When  $S \subseteq \mathbb{N}^n$ , the ring  $\mathbb{k}[S]$  is a subalgebra of the polynomial ring  $\mathbb{k}[t_1, \dots, t_n]$ . More generally, if  $S \subseteq \mathbb{Z}^n$ , then  $\mathbb{k}[S]$  is naturally a subalgebra of the Laurent polynomial ring

$$\mathbb{k}[t_1^{\pm 1}, \dots, t_n^{\pm 1}].$$

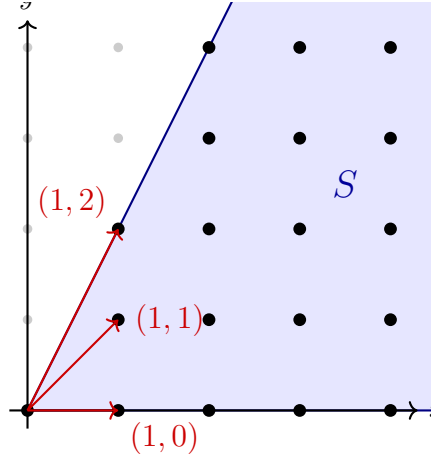
**Example 1.26.** Let

$$S = \mathbb{N}\{(1, 0), (1, 1), (1, 2)\} \subseteq \mathbb{Z}^2.$$

Then

$$\mathbb{k}[S] = \mathbb{k}[t_1, t_1 t_2, t_1 t_2^2] \subseteq \mathbb{k}[t_1, t_2].$$

Notice that  $(1, 1) = \frac{1}{2}((1, 0) + (1, 2)) \in \text{pos}\{(1, 0), (1, 2)\}$ , but  $(1, 1)$  cannot be written as a sum of other nonzero elements in  $S$ , so it is a necessary Hilbert-basis element.



**Proposition 1.27.** *Let  $S$  be an affine semigroup. Then:*

- (1)  $\mathbb{k}[S]$  is an integral domain.
- (2)  $\dim \mathbb{k}[S] = \text{rank}(S)$ . If  $S = \sigma \cap \mathbb{Z}^n$ , then  $\dim \mathbb{k}[S] = \dim \sigma$ .
- (3)  $\mathbb{k}[S]$  is a finitely generated  $\mathbb{k}$ -algebra.
- (4) If  $S$  is saturated, then  $\mathbb{k}[S]$  is integrally closed.

*Proof.* Since  $\mathbb{k}[S]$  embeds into the Laurent polynomial ring

$$\mathbb{k}[t_1^{\pm 1}, \dots, t_n^{\pm 1}],$$

it is an integral domain.

The dimension statement follows because the fraction field of  $\mathbb{k}[S]$  is generated by the characters corresponding to the group  $\mathbb{Z}S$ . Therefore the transcendence degree of  $\text{Frac}(\mathbb{k}[S])$  over  $\mathbb{k}$  is  $\text{rank}(S)$ , and hence

$$\dim \mathbb{k}[S] = \text{rank}(S).$$

If  $S = \mathbb{N}\{m_1, \dots, m_r\}$ , then

$$\mathbb{k}[S] = \mathbb{k}[t^{m_1}, \dots, t^{m_r}],$$

which proves it is finitely generated.

Finally, the equivalence between saturation of  $S$  and normality of  $\mathbb{k}[S]$  is the standard normality criterion for affine semigroup rings. Thus, if  $S$  is saturated, then  $\mathbb{k}[S]$  is integrally closed.  $\square$

*Remark 1.28.* The last point is the algebraic heart of the toric dictionary:

$$S \text{ saturated} \iff \mathbb{k}[S] \text{ normal}$$

Affine toric varieties are exactly spectra of saturated affine semigroup rings.



## Presentations by Binomial Ideals

Let

$$S = \mathbb{N}\{m_1, \dots, m_r\} \subseteq \mathbb{Z}^n.$$

There is a surjective  $\mathbb{k}$ -algebra homomorphism

$$\varphi : \mathbb{k}[T_1, \dots, T_r] \longrightarrow \mathbb{k}[S]$$

given by

$$T_i \longmapsto t^{m_i}.$$

By the First Isomorphism Theorem,

$$\mathbb{k}[S] \cong \frac{\mathbb{k}[T_1, \dots, T_r]}{\ker \varphi}.$$

**Example 1.29.** Let

$$S = \mathbb{N}\{(1, 0), (1, 1), (1, 2)\}.$$

Let

$$m_1 = (1, 0), \quad m_2 = (1, 1), \quad m_3 = (1, 2).$$

The map is

$$\varphi : \mathbb{k}[T_1, T_2, T_3] \longrightarrow \mathbb{k}[S]$$

with

$$T_1 \longmapsto t_1, \quad T_2 \longmapsto t_1 t_2, \quad T_3 \longmapsto t_1 t_2^2.$$

Notice that

$$m_1 + m_3 = 2m_2,$$

which implies that

$$\varphi(T_1 T_3) = t_1^2 t_2^2 = \varphi(T_2^2).$$

Thus,

$$T_1 T_3 - T_2^2 \in \ker \varphi.$$

In fact, it is true that  $\ker \varphi = (T_1 T_3 - T_2^2)$ , so

$$\mathbb{k}[S] \cong \frac{\mathbb{k}[T_1, T_2, T_3]}{(T_1 T_3 - T_2^2)}.$$

$$\begin{array}{ccc} \mathbb{k}[T_1, T_2, T_3] & \xrightarrow{\varphi} & \mathbb{k}[S] \\ \downarrow & \nearrow \sim & \\ \mathbb{k}[T_1, T_2, T_3]/(T_1 T_3 - T_2^2) & & \end{array}$$

**Theorem 1.30** (Affine semigroup rings have binomial presentations). *Let  $S$  be an affine semigroup generated by  $m_1, \dots, m_r$ . Then*

$$\mathbb{k}[S] \cong \frac{\mathbb{k}[T_1, \dots, T_r]}{I_S},$$

where  $I_S$  is a binomial ideal.

*Proof.* The map

$$\varphi : \mathbb{k}[T_1, \dots, T_r] \rightarrow \mathbb{k}[S], \quad T_i \mapsto t^{m_i},$$

maps monomials to monomials. For  $\alpha = (\alpha_1, \dots, \alpha_r) \in \mathbb{N}^r$ , we have

$$\varphi(T^\alpha) = t^{\alpha_1 m_1 + \dots + \alpha_r m_r}.$$

Thus  $\varphi(T^\alpha) = \varphi(T^\beta)$  when

$$\sum_{i=1}^r \alpha_i m_i = \sum_{i=1}^r \beta_i m_i.$$

The kernel is generated by the binomials  $T^\alpha - T^\beta$ . □

## Multigrading and Affine Toric Geometry

The semigroup ring  $\mathbb{k}[S]$  is naturally graded by the group  $\mathbb{Z}S$ :

$$\deg(t^m) = m.$$

Under the presentation

$$\mathbb{k}[S] \cong \mathbb{k}[T_1, \dots, T_r] / I_S,$$

this corresponds to setting

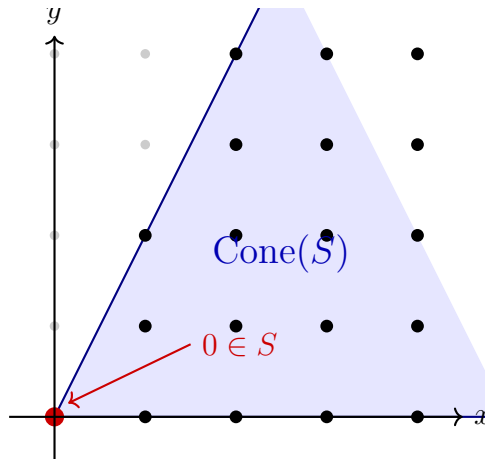
$$\deg(T_i) = m_i.$$

Geometrically, the presentation

$$\mathbb{k}[S] \cong \frac{\mathbb{k}[T_1, \dots, T_r]}{I_S}$$

gives a closed embedding of the affine toric variety into affine space:

$$\mathrm{Spec} \mathbb{k}[S] \hookrightarrow \mathrm{Spec} \mathbb{k}[T_1, \dots, T_r] \cong \mathbb{A}^r.$$



*Remark 1.31.* The final note from the lecture is that the minimal graded free resolution of binomial ideals is far from understood. Thus, even though affine semigroup rings have explicit binomial presentations, the homological algebra of their defining ideals can be highly nontrivial.

## 2 Lecture 2- May 26, 2026

### Affine Semigroups and Hilbert Series

Let  $G \subseteq \mathbb{R}^n$  be a rational polyhedral cone defined by:

$$G = \{x \mid Ax \geq 0\} \subseteq \mathbb{R}^n, \quad A \in \mathbb{Z}^{m \times n}, \quad m \geq n$$

Assume that  $G$  is strongly convex, meaning  $G \cap (-G) = \{0\}$ .

We define the affine semigroup  $S$  as:

$$S = G \cap \mathbb{Z}^n$$

The cone  $G$  can also be described as the positive hull of  $S$ :

$$G = \text{pos}(S) = \left\{ \sum \lambda_i s_i \mid \lambda_i \geq 0, s_i \in S \right\}$$

We assume  $\dim(G) = n$ .

Let  $K$  be a field with  $\text{char}(K) = 0$ . The semigroup ring is given by:

$$K[S] = K[G \cap \mathbb{Z}^n]$$

This ring can be graded in two ways:

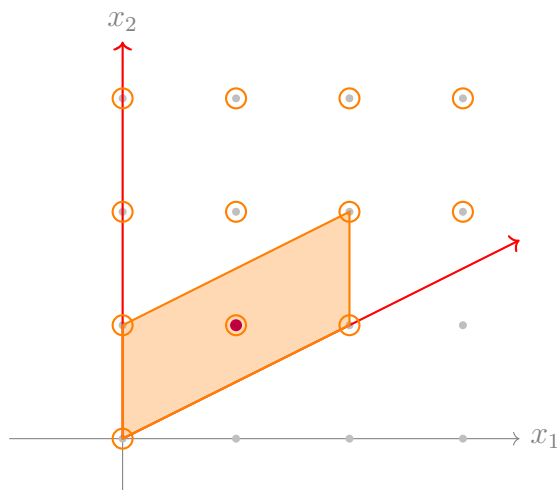
1. **Fine grading:**  $K[S]$  is graded by  $\mathbb{Z}^n$ , where  $\deg(t^m) = m$ .
2. **Coarse grading:**  $K[t^m \mid m \in S]$  is graded by  $\mathbb{Z}$ , where  $\deg(t^m) = \sum m_i$ .

**Definition 2.1** (Hilbert Series). Using the fine grading, the Hilbert series of  $K[S]$  is given by:

$$\text{Hilb}(K[S]; z) = \sum_{m \in S} z^m$$

**Example 2.2.** Consider the cone in  $\mathbb{R}^2$  defined by the inequalities  $x_1 \geq 0$  and  $-x_1 + 2x_2 \geq 0$ . The rays of this cone are generated by the columns of the matrix:

$$\begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix}$$



Notice that the subsemigroup generated by the rays does not cover the entire intersection  $G \cap \mathbb{Z}^2$ :

$$\mathbb{Z} \left[ \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix} \right] \subseteq \mathbb{Z}^2$$

It's not covering the whole thing; the point corresponding to  $z_1 z_2$  (which is  $(1, 1)$ ) is missing from the ray generators.

The generating function for just the rays would be:

$$\begin{aligned} H \left( \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix} \right) &= \left( \frac{1}{1 - z_2} \right) \times \left( \frac{1}{1 - z_1^2 z_2} \right) \\ &= (1 + z_2 + z_2^2 + \dots) \times (1 + z_1^2 z_2 + \dots) \\ &= 1 + z_2 + z_1^2 z_2 + z_2^2 + z_1^2 z_2^2 + \dots \end{aligned}$$

To get the full Hilbert series, we must account for the missing point  $(1, 1)$  inside the fundamental parallelogram. Thus, the correct Hilbert series is:

$$\begin{aligned} \text{Hilb}(K[S]; z_1, z_2) &= \frac{1}{(1 - z_2)(1 - z_1^2 z_2)} + \frac{z_1 z_2}{(1 - z_2)(1 - z_1^2 z_2)} \\ &= \frac{1 + z_1 z_2}{(1 - z_2)(1 - z_1^2 z_2)} \end{aligned}$$

Now, assume  $G$  is simplicial, meaning  $G = \text{pos}\{v_1, \dots, v_n\}$  where the  $v_i$  are linearly independent and the gcd of the entries of each  $v_i$  is 1. We define the fundamental parallelogram as:

$$\text{Par}\{v_1, \dots, v_n\} = \left\{ \sum \lambda_i v_i \mid 0 \leq \lambda_i < 1 \right\}$$

**Theorem 2.3.** *If  $G$  is simplicial, the Hilbert series is given by:*

$$\text{Hilb}(K[S]; z) = \frac{\sum_{m \in \text{Par} \cap \mathbb{Z}^n} z^m}{\prod_{i=1}^n (1 - z^{v_i})}$$

*Remark 2.4.* Refer to the graph on the previous page to explain how points in the fundamental parallelogram shift the ray-generated points to cover the entire semigroup cone.

## Cohen-Macaulay Property and Free Resolutions

Let  $G = \text{pos}\{v_1, \dots, v_e\}$  with  $e \geq n$  being finite. We have:

$$K[t^{v_1}, \dots, t^{v_e}] \subseteq K[S] \quad (\text{finite})$$

The polynomial ring  $K[T_1, \dots, T_e]$  (which is not finite) maps surjectively onto  $K[S]$  via  $T_i \mapsto t^{v_i}$ . This implies that  $K[S]$  is a finite  $K[T]$ -module.

Taking  $Q = K[T]$ , we can construct a graded free resolution of  $Q$ -modules:

$$0 \rightarrow F_p \rightarrow \dots \rightarrow F_1 \rightarrow F_0 \rightarrow K[S] \rightarrow 0$$

This yields a rational expression for the Hilbert series:

$$\text{Hilb}(K[S]; z) = \frac{P(z)}{\prod_{i=1}^e (1 - z^{v_i})}$$

**Theorem 2.5** (Hochster).  $K[S]$  is Cohen-Macaulay (CM). Recall that  $K[S]$  is CM if and only if one (equivalently every) homogeneous system of parameters (hsop) is a regular sequence.

Under the coarse grading, let  $\theta_1, \theta_2, \dots, \theta_n \in K[S]$  be a homogeneous system of parameters. If  $K[S]$  is a finite module over  $K[\theta_1, \dots, \theta_n]$ , then the hsop is a regular sequence if and only if  $K[S]$  is a free module over  $K[\theta_1, \dots, \theta_n]$ .

**Theorem 2.6.** If  $\theta_1, \dots, \theta_n$  are a regular sequence, then:

$$\text{Hilb}(K[S]; z) = \frac{\sum z^{\deg(\eta_i)}}{\prod_{i=1}^n (1 - z^{\deg(\theta_i)})}$$

where  $\eta_1, \dots, \eta_s$  are a basis for  $K[S]$  as a  $K[\theta_1, \dots, \theta_n]$ -module.

*Remark 2.7.* A homogeneous system of parameters exists in  $K[S]$ , existing in degree 1 by Noether Normalization. Here, the semigroup  $S$  needs to be saturated.

## Local Cohomology and the Canonical Module

**Proof Strategy:** Stanley and Ishida compute the local cohomology to prove these properties.

Let  $I \subseteq K[S]$  be the ideal  $I = \langle t^m \mid m \in S, m \neq 0 \rangle$ . Define the left exact additive functor:

$$L_I(M) = \{u \in M \mid I^n u = 0 \text{ for } n \gg 0\}$$

The local cohomology modules are defined as the right derived functors:

$$H_I^i(M) = R^i L_I(M)$$

We can compute this using the Čech / Ishida complex.

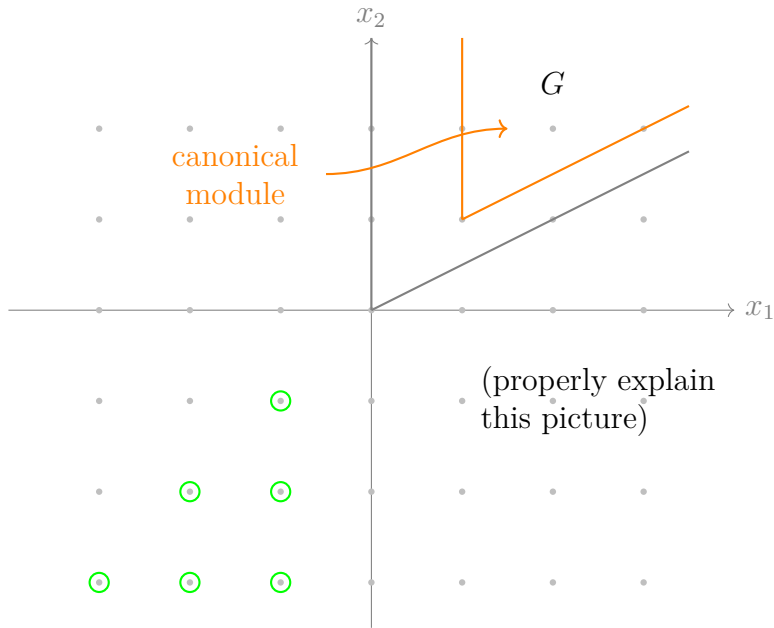
**General picture:**

$$H_I^i(K[S]) = 0 \text{ for } 0 \leq i < n$$

For the top cohomology:

$$H_I^n(K[S]) = K\{t^m \mid A(-m) > 0, m \in \mathbb{Z}^n\}$$

which is supported on  $Am < 0$ . Geometrically, this corresponds to lattice points in the strict interior of the opposite cone,  $-\text{int}(G)$ .



**Definition 2.8** (Canonical Module). The canonical module  $\Omega(K[S])$  is defined algebraically by:

$$\Omega(K[S]) = K\{t^m \mid Am > 0\} = K\{\text{int}(G)\}$$

This implies that  $\Omega(K[S])$  is an ideal of  $K[S]$ . If  $\Omega(K[S]) \sim K[S]$  as modules, it implies that  $K[S]$  is a Gorenstein ring.

For the running example, the generator of the interior of the cone is  $t^{(1,1)}$ . Thus:

$$\Omega(K[S]) = t^{(1,1)} \cdot K[S] \implies \text{Gorenstein}$$

*Next  $\rightarrow$  Ehrhart Polynomials.*